Reiwa 6 (2024)

Date submitted: 6/10/2024

Fundamental Labs I Report

(School of Science and Engineering, College of Engineering Systems)

- ☐ 7 System control engineering basic labs
- ☐ 8 Basics of linear systems using operational amplifier
- ☐ 9 Diodes and transistors
- ☑ 10 Basics of logic circuits and computers
- ☐ 11 DC motor manufacturing and control
- ☐ 12 Mechanisms and mechanical elements

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Lab partners: (1) Chong Shao (2) _____
(3) _____ (4) _____
(5) _____ (6) _____

	1st week	2nd week	Report submission
Lab date	Month/Day	Month/Day	Received stamp
Lab participant's stamp with seal Non-participants leave blank	24.12.09	24.12.16	
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Fundamental Labs Chapter 10 Report: Computer and Logic Operation

Purpose of lab To understand the basics of computer hardware operation and memory storage using logic gates, combination circuits, flip-flops, shift resistors, and asynchronous and synchronous counters

Week 1

Introduction and Purpose

A logic circuit handles signals that have two types of values (1 and 0). 1 and 0 correspond to two logical values (true and false). A system consisting of operations defined by two logical values is referred to as logical algebra. When logical algebra is used, it is easy to describe the properties of a logic circuit. A combination circuit is a logic circuit in which the output is dependent on combinations of the input value at a given time. In contrast, a logic circuit in which the output is dependent upon the internal state of the circuit itself is called a sequential circuit. A flip-flop is used to maintain the internal state of the circuit.
The purpose of this lab is to understand the operation of a combination circuit when making simple circuits, and to understand the operations of a flip-flop.

Assignment 1.1 (Lab 1.2: 3-8 decoder configuration)

Experimental details

In Lab 1.2, we configured a 3-8 decoder, a decoder circuit with three inputs and eight outputs using three binary digits as the input.

Truth table and logical expressions

The truth table for a three-input eight-output decoder is the following below in Figure 1:

A	B	C	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

Figure 1 Truth table for a three-input eight-output decoder

From the truth table in Figure 1, the logic expression in Boolean expressions can be defined as

$$Y_0 = A'B'C'$$

$$Y_1 = A'B'C$$

$$Y_2 = A'BC'$$

$$Y_3 = A'BC'$$

$$Y_4 = AB'C'$$

$$Y_5 = A'B'C'$$

$$Y_6 = ABC'$$

$$Y_7 = ABC$$

From the above Boolean expressions, the implementation of 3 to 8 decoder circuit can be done with the usage of three NOT gates & 8-three input AND gates.

Circuit diagram

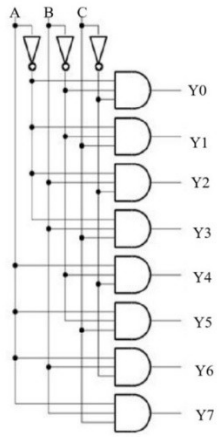


Figure 2 3-8 encoder logic gate circuit diagram

How The Circuit Works

In the above circuit in Figure 2, the three inputs can be decoded into 8 outputs, where every output represents one of the midterms of the three input variables.

The 3 inverters in the above logic circuit will provide the complement of the inputs & each one of the AND gates will generate one of the midterms.

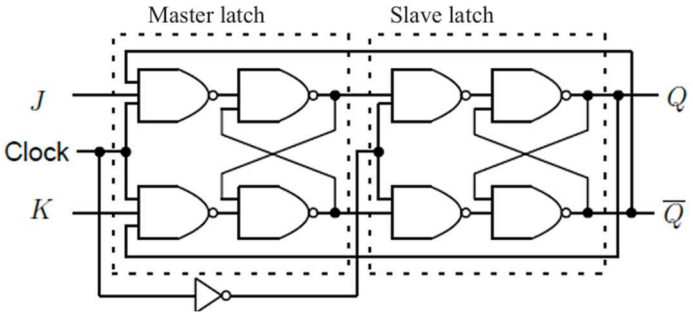
This kind of decoder mainly used to decode any 3-bit code & generates eight outputs, equivalent to 8 different combinations for the input.

Assignment 1.2 (Lab 1.6 JK flip-flop operating principle confirmation)

Experimental details

In this lab, we set both the inputs J and K to 1, and the hardwire so that a 1-kHz clock pulse is supplied to the input T. Simultaneously we observed the input T and the output Q at this time with two oscilloscope channels, and recorded the waveform.

The master-slave type JK flip-flop circuit diagram and the truth table are in the figure below. CK = Clock



J^n	K^n	CK	Q^{n+1}
0	0	⎍	Q^n
0	1	⎍	0
1	0	⎍	1
1	1	⎍	$\overline{Q^n}$

Figure 3 Master-slave type JK flip-flop and truth table

Waveforms (Input T and Output Q waveforms)

We observed the input T and the output Q at this time with two oscilloscope channels and recorded the waveform for each. The input T and output Q waveforms with timing chart and arrows showing causal relationship are in Figure 4 below. Figure 4 shows changes in output with changes in input along a time axis, which is referred to as a timing chart. Arrows have been added to clearly indicate where there is a causal relationship between changes.

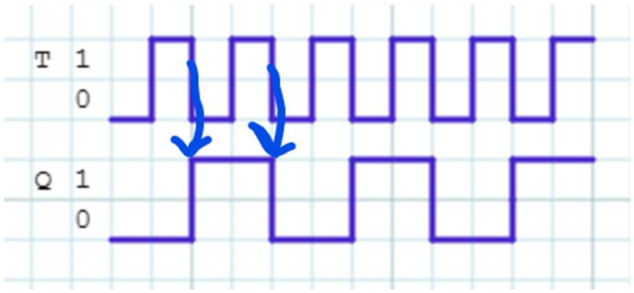


Figure 4 JK flip-flop timing chart

The relationship between the two waveforms

1) When the input T was changed from 1 to 0, the output Q will change. However, when T was changed from 0 to 1, the value of Q will not change. The truth table for the relationship between the two waveform is shown in Table 1 in number 3
2) The frequency of Q is one-half of the frequency of T.
3) The truth table for the relationship in number (1) is the following:

J	K	Clock	T	Q
1	1	⎍	0	0

1	1		1	0
1	1		0	1
1	1		1	1
1	1		0	0

Table 1 Truth Table for the relationship between T and Q for J =1 and K=1 in a master-slave JK-flip-flop

When both J and K are 1, and the clock input is in the "High" state, the output toggles its state.

Assignment 1.3 (Lab 1.8 Observation of edge trigger system operating principle)

Experimental Details

In the circuit in Figure 5, the two previous stage SR flip-flops work to cut out the clock edge. We observed this operation using the following method:

a) Hardwire a 1-kHz clock pulse to the clock input b) Hardwire the output Q of the JK flip-flop observed in Lab 1.6 to the input D. The timing chart of the clock input and input D becomes exactly as that in Figure 6. We then added the output Q, A1 and A2 point signals, and B1 and B2 point signals to this timing chart. We then indicated the causal relationships in the changes. Furthermore, we observed the signals at points A1, A2, B1, and B2 using the oscilloscope. In this case, we always observed the input D with channel 1, and also synchronized it with channel 1. We observed signal at points A1, A2, B1, and B2 using Channel 2, and ascertain the time relationship with the input D.

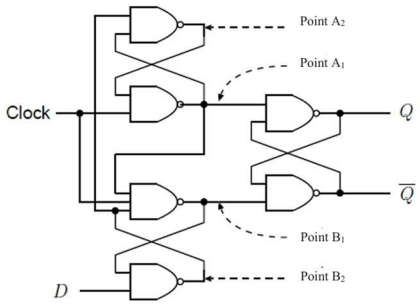


Figure 5 Edge trigger-type D flip-flop

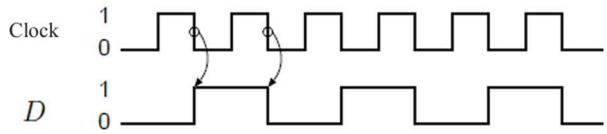


Figure 6 D flip-flop timing chart for inputs Clock and D

Resulting waveform with outputs A1, B1, B2, and Q

The timing chart for the D flip-flop with the inputs CK(Clock) and D, and the outputs point A1, point B1, point B2, and Q are in Figure 7 below.

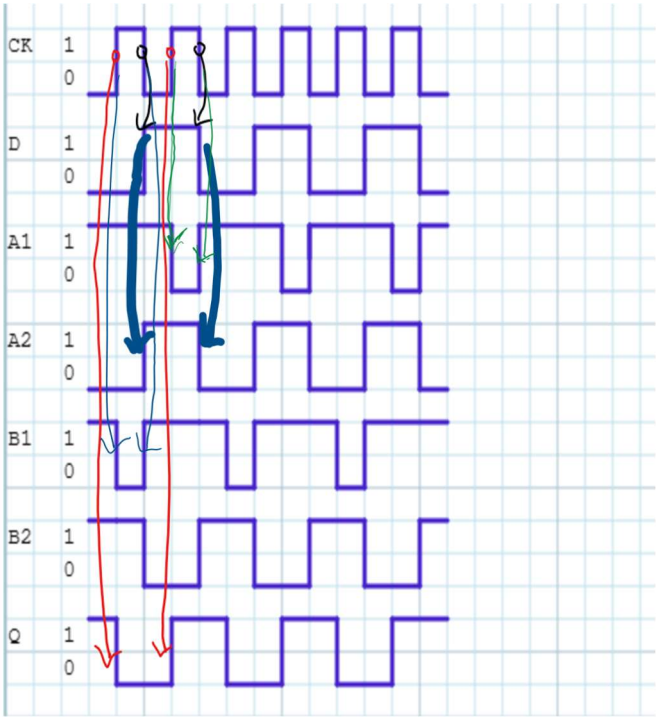


Figure 7 Positive edge type D-flip flop timing chart, with CK as clock.

Positive edge type D -Flip-Flop (Positive Edge Type D-FF)

a) Positive edge

Positive edge-triggered flip-flop is the type of edge-triggered flip-flop whose output changes its state only on the rising edge (edge that goes from low to high) of the clock pulse, i.e. when the clock rises from 0 to 1. In a positive edge triggered flip flop, the inputs are accepted and stored only when the clock pulse goes from low (0) to high (1), i.e. on the rising edge of the clock pulse. This stored value is then available on the outputs.

From Figure 7 and the red arrow showing the causal relationship between Clock and output Q, we can see that the state/value of output Q only changes when the clock rises from 0 to 1.

b)D-FF

In a positive edge-triggered D flip-flop, the relationship between the input D and the output Q is characterized by the following behaviour:

1) The output Q assumes the same logic level as the input D at the positive (rising) edge of the clock signal12.

2) Once the positive edge of the clock has passed, the output Q remains stable and ignores any changes in the D input until the next positive clock edge2.
3) The characteristic equation of a D flip-flop is Q(n+1) = D, which means the next state of the output (Q) will be equal to the current state of the input (D)13.
4) If D is low (0) at the positive clock edge, Q will be set to 0. If D is high (1) at the positive clock edge, Q will be set to 11.
5) The D flip-flop acts as a delay element, as it "delays" the input signal by one clock cycle before it appears at the output

Additional question of Assignment 3.3

If we separate the circuit into three SR-Flip Flops as in Figure 8,

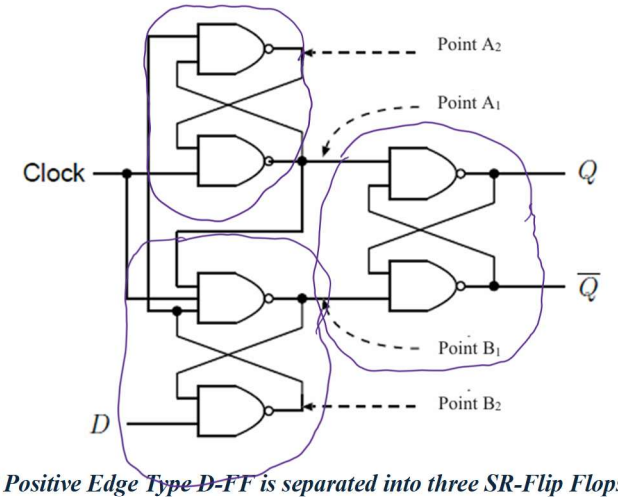


Figure 8 Positive Edge Type D-FF is separated into three SR-Flip Flops

1) When CK=0, A1, B1, and Q are all 1.
   When D=0, A1 is 1. For B1 and Q, initially, when both CK=0 and D=0, the values of B1 and Q are zero. However, during D=0, when CK raises to 1, both B1 and Q fall to 0.
   When D=1, A1 is initially 1 and B1 is initially 0 until while D=1 the Clock rises from 0 to 1. Then A1 becomes 0 and B1 rises to 1.
2) The SR Flip-Flops:

1. SR Flip-Flop 1 (A):
   - Inputs: S_A, R_A
   - Produces intermediate output A1 and A2.
2. SR Flip-Flop 2 (B):
   - Inputs: S_B, R_B
   - Produces intermediate output B1 and B2.
3. SR Flip-Flop 3 (Q):
   - Inputs: S_Q, R_Q
   - Produces the final output Q and $\bar{Q}$.

Step-by-Step Operation:

1. Input Analysis:

- The data input D determines whether Q should be set to 1 (if D=1) or reset to 0 (if D=0).
- The clock signal CLK triggers changes in the outputs only on the positive edge.

2. Behaviour of SR Flip-Flops:

1. SR Flip-Flop 1 (A):
    - This flip-flop responds directly to the D input and the clock signal.
    - At the rising edge of the clock:
        - If D=1, S_A=1 and R_A =0, causing A1=1 (set state).
        - If D=, S_A=0 and R_A =1, causing A1=0 (reset state).
2. SR Flip-Flop 2 (B):
    - This flip-flop is clocked by CLK and controlled by the outputs of Flip-Flop 1 (A1 and A2).
    - At the rising edge of the clock:
        - If A1=1, S_B =1 and R_B=0, causing B1=1.
        - If A1=0, S_B =0 and R_B =1, causing B1=0.
3. SR Flip-Flop 3 (Q):
    - The final flip-flop produces Q, which holds the output value.
    - At the rising edge of the clock:
        - If B1=1, S_Q=1 and R_Q =0, causing Q=1.
        - If B1=0, S_Q=0 and R_Q =1, causing Q=0.

3. Overall Behaviour:

- The flip-flops are chained so that the value of D propagates through the stages and is captured by Q at the rising edge of the clock.
- Between clock edges, the intermediate states (A1, A2,B1,B2) remain constant, ensuring stability.

Thus the truth table of each SR FF are respectively in Figure 9 below. 9 The truth table of SR FF1, SR FF2, and SR FF3 are the same, i.e. this truth table.

S^n	R^n	Q^{n+1}	$\overline{Q^{n+1}}$
0	0	Q^n	$\overline{Q^n}$
0	1	0	1
1	0	1	0
1	1	1	1

Figure 9 The truth table of SR FF1, SR FF2, and SR FF3 are the same, i.e this truth table

3) The circuit shown separates the D flip-flop into three SR flip-flops, where each flip-flop contributes to the overall behavior. The outputs Q and of the D flip-flop depend on the combined functionality of these SR flip-flops.

Truth table for the positive edge-triggered D flip-flop:

Using the design, we derive the overall truth table for the D flip-flop. The truth table is in Figure 10.

Inputs		Outputs		
D		**Q**	$\overline{\mathbf{Q}}$	**Comments**
CLK				
1	↑	1	0	SET (stores a 1)
0	↑	0	1	RESET (stores a 0)

Figure 10 truth table for positive edge–triggered D flip-flop

References

1) 3 to 8 Line Decoder : Designing Steps & Its Applications

Url: https://www.elprocus.com/designing-3-line-to-8-line-decoder-demultiplexer/

2) 3 to 8 Decoder and truth table of 3 to 8 decoder.

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3) JK Flip Flop Circuit, Truth Table, Limitations, and Uses
Url: https://www.electronicsforu.com/technology-trends/learn-electronics/jk-flip-flop-circuit-truth-table-limitations-applications
4) Edge-Triggered Flip-Flop
Url: https://www.tutorialspoint.com/digital-electronics/edge-triggered-flip-flop.htm
5) D Flip-Flop and Edge-Triggered D Flip-Flop With Circuit diagram and Truth Table
Url: https://www.electroniclinic.com/d-flip-flop-and-edge-triggered-d-flip-flop-with-circuit-diagram-and-truth-table/

6) Petzold, Charles. *The Code: The Hidden Language of Computer Hardware and Software*, 2nd edition, 2023

Week 2

Introduction and Purpose of the Lab

The purpose of this lab is to make a shift resistor and counter as a simple example of a sequential circuit and to understand its operation. A sequential circuit is a logic circuit in which the circuit output is determined not only by the present output but depends on the past to present input series. Past input series are remembered in the internal state of the circuit. It is possible to use flip-flops in the internal state memory.

A) Asynchronous counter configuration

Mod-K Asynchronous counter

(The explanation for the process of mod-K asynchronous counter and control method for the state transition from K-1 to 0 with processes (a), (b), (c) and (d) is in *Fundamental Labs Textbook of College of Engineering Systems University of Tsukuba, October 2024 edition* pages 104-105.)

Use n number of JK flip-flops, and think about making a mod-K asynchronous counter that is not an exponent of 2, but counts from 0 to K-1 for general K ($2^{n-1} < K < 2^n$). Write the ith flip-flop as FFi, and write the output Q of FFi as Qi. The numerical value at which n number of Qi (Q0, Q1, . . ., Qn Q1, . . ., Qn 1) are expressed, or the (Q0, 1) when Qn-1 is seen as the highest-order bit and Q0 is seen as the lowest-order bit, is called the shift resistor state. We find that to make a mod-K asynchronous counter, signals supplied to the T terminal of each flip-flop may be controlled so that the shift resistor in state K-1 returns to state 0 when the Kth clock pulse is introduced.

(1) Control method for state transition from K-1 to 0:
In general, in state K-1, there are four combinations as follows for the two outputs of bit Qi (1 i n) and bit Qi-1 just one before. The following steps may be taken in each case so that bit Qi is 0 for the following states. In order to transition the shift resistor state from K-1 to 0, perform this step for all flip-flops. However, when Qi is a lowest-order bit, Qi-1 is an input clock, and is considered to be 1.
(a) When (Qi-1 , Qi)=(0, 0) With respect to FFi, even if nothing is done, Qiremains 0. 104
(b) When (Qi-1 , Qi) = (1, 1) With respect to FFi, even if nothing is done, Qi is inverted to 0.
(c) When (Qi-1 , Qi) = (0, 1) Insert forced pulse into the T terminal of FFi, and invert Qi to 0. The forced inserted pulse needs to be limited to state K – 1, so gate the input clock pulse (specifically use AND gate) and set Qi to 0 with a signal that

reaches 1 only in state K-1. Take this signal and the OR of the original signal Qi-1, and supply to the T terminal of FFi.
(d) When (Qi-1 , Qi) = (1, 0) When Qi-1 changes with the next input clock pulse and reaches 0, Qi inverts and reaches 1, so 0 is conferred upon the input J of FFi; when Qi-1 changes to 0, 0 is set for Qi (that is, it does not change). However, even earlier than the time that the input of the T terminal of FFi reaches 1, there is a need to settle the input J of FFi at 0. When delays in later-stage changes in FF are considered, a signal that reaches 0 from when one state K-1 before is made, and is provided for the input J.

Assignment 2.1 (Lab 2.4 Decimal asynchronous counter configuration)

Experimental details

We configured a decimal asynchronous counter that counts from 0 as an example of mod-K asynchronous counter (K=10) on the training device and confirmed operation. As decimal is mod-10 (K=10), we make a mod-10 asynchronous counter that counts from 0 to K-1 (10-1=9). We used n number of JK flip-flops such that $2^{n-1} < K < 2^n$. For K=10, the value of n is 4, so we used 4 JK flip-flops. For a decimal counter or mod-10 asynchronous counter, the lowest-order bit is Q0 and the highest-order bit is Q3. The bit values of state 9 (K-1=10-1=9) ((Q3, Q2, Q1, Q0) = (1001)) and state 0 ((Q3, Q2, Q1, Q0) = (0000)) are as in Figure 11. Take care that bit Qi is inverted when bit Qi-1, i.e., the output of the previous stage FFi-1, has changed from 1 to 0. Consider four-bit combinations in state 9:

(Q2, Q3) = (0, 1), process (c) for T terminal of FF3

(Q1, Q2) = (0, 0), process (a) for T terminal of FF2

(Q0, Q1) = (1, 0), process (d) for T terminal of FF1

(CK, Q0) = (1, 1), process (b) for T terminal of FF0 may each be performed.

The processes (a), (b), (c) and (d) in the control method for state transition from K-1 to 0 for mod-K asynchronous counter are in *Fundamental Labs Textbook of College of Engineering Systems University of Tsukuba, October 2024 edition* pages 104-105.

	Q_0	Q_1	Q_2	Q_3
State 9	1	0	0	1
State 0	0	0	0	0
(Q_{i-1}, Q_i)	(b)	(d)	(a)	(c)

Figure 11 Decimal asynchronous counter state transition

To make a decimal asynchronous counter, signal supplied to the T terminal of each flip-flop may be controlled so that the shift resistor in state 9 (K-1=10-1) returns to state 0

when the 10th (K=10) clock pulse is introduced. Thus, the decimal counter is reset to 0 when the output count hits the decimal value of 10 or in binary 1010.

The decimal asynchronous counter counts from 0000 to 1001. Rest of the states are invalid. To design the combinational circuit of valid states, following truth table below:

Q0	Q1	Q2	Q3	Reset
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0

From the truth table we can determine the reset logic. This will be given to the reset inputs of the counter so that as soon as count 1010 reaches, the counter will reset. Thus the counter will count from 0000 to 1001.

Here we configured the circuit and confirmed the operation. The circuit diagram in Figure 12.

Circuit diagram

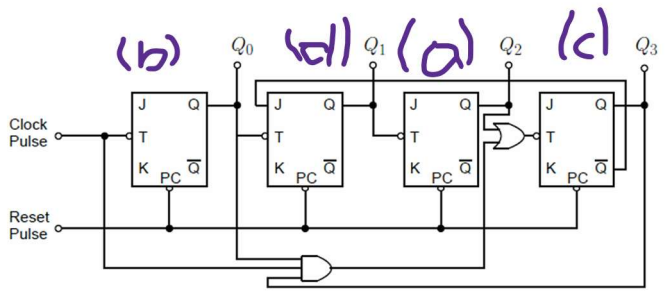


Figure 12 Decimal Asynchronous counter

FF0 undergoes process (b), FF1 goes through process (d), FF2 goes through process (a), and FF3 goes through process (c) of the mod-K asynchronous counter transition from state of K-1 to K (reset or 0), with K=10.

Result (Waveform)

The waveforms of the decimal asynchronous counter in Lab 2.4 are in Figure 13. The waveforms are of the Clock, outputs Q0, Q1, Q2, and Q3, J1, the J input of the JK flip-flop outputting Q1, and T3, the input T of the JK flip-flop outputting Q3. The arrows show the causal relationship between the waveforms.

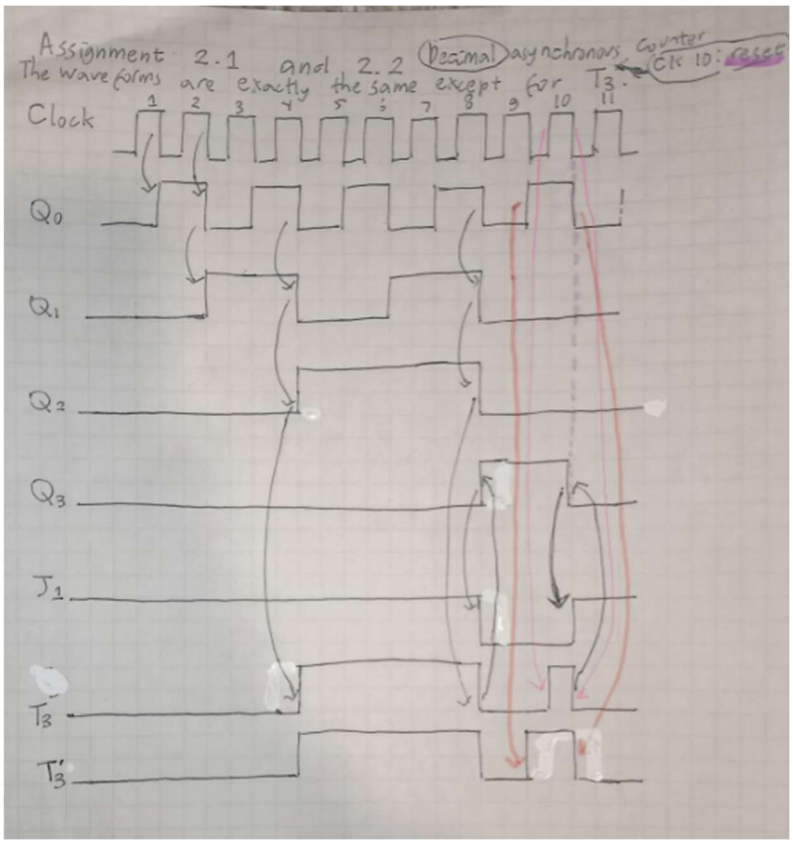


Figure 13 Waveforms for the Decimal Asynchronous Counter in Assignments 2.1 and 2.2

The counter resets when the clock pulse reaches 10. During the 10th clock pulse

(1) Q0 and Q3 falls from 1 to 0 when the 10th clock pulse falls from 1 to 0. In the 11th clock pulse, the values of Q0 and Q3 remain the same

J1, the J input of the JK flip-flop outputting 1, has a load to prevent Q1 to rise to 1. While T3 has a load to force Q3 to rise to 1.

Assignment 2.2 (Lab 2.5 Decimal asynchronous counter configuration)

We configured a decimal asynchronous counter that is slightly more simplified than the one in Lab 2.4 (Figure 1). The only difference is that in this simplified circuit, the Clock pulse is not inputted to the AND circuit. Now the AND circuit has inputs Q0 (the output of FF(flip-flop) 0) and Q3 (the output of FF3) instead of Clock, Q0, and Q3.

The circuit diagram of the simplified decimal asynchronous counter is in Figure 14.

Circuit diagram

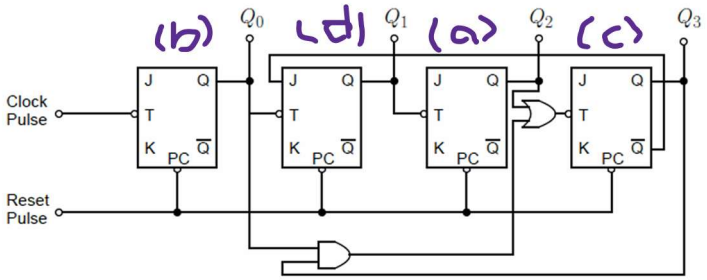


Figure 14 Decimal Asynchronous counter (simplified) in Lab 2.5

Waveform

The waveforms for Assignment 2.1 and 2.2 are the same, except for T3. In Figure 14, T3 from Assignment 2.2 is labelled as T'3 (T prime 3). The arrows show a causal relationship between the waveforms.

Merit of configuration of the decimal asynchronous circuit in Assignment 1 (Figure 12) and Assignment 2 (Figure 14)

Assignment 2.1: Compared to the decimal asynchronous counter configuration in assignment 2.2 (Lab 2.5), the decimal asynchronous counter configuration in assignment 2.1 uses more general method, so it is more adaptable and can be used everywhere.

Assignment 2.2: Compared to the decimal asynchronous counter configuration in assignment 2.1, the decimal asynchronous counter configuration in assignment 2.2 is simpler and uses less wire, so is suitable for mass production.

Assignment 2.3 Lab 2.6 mod-13 (K) asynchronous counter configuration

Experimental details We designed and configured a mod-13 asynchronous counter, which is a mod-K asynchronous counter with K=13. We also made a timing chart.

Circuit diagram

The decimal asynchronous counter counts from 0000 (decimal value 0) to 1101 (decimal value 13). The shift resistor in state 12 (K-1=13-1) returns to state 0 when the 13th (K=13) clock pulse is introduced. Thus, the decimal counter is reset to 0 when the output count hits the decimal value of 13 or in binary 1101.

We made a mod-13 asynchronous counter that counts from 0 to K-1 (13-1=9). We used n number of JK flip-flops such that $2^{n-1} < K < 2^n$. For K=13, the value of n is 4, so we used 4 JK flip-flops. For a mod-13 asynchronous counter, the lowest-order bit is Q0 and the highest-order bit is Q3. The bit values of state 12 (K-1=13-1=13) ((Q3, Q2, Q1, Q0) = (1100)) and state 0 ((Q3, Q2, Q1, Q0) = (0000)) are as in Figure 15. Take care that bit Qi is inverted when bit Qi-1, i.e., the output of the previous stage FFi-1, has changed from 1 to 0.

Consider four-bit combinations in state 12:

(Q2, Q3) = (1, 1), process (b) for T terminal of FF3

(Q1, Q2) = (0, 1), process (c) for T terminal of FF2

(Q0, Q1) = (0, 0), process (a) for T terminal of FF1

(CK, Q0) = (1, 0), process (b) for T terminal of FF0 may each be performed.

State (Clock)	Q0 (2^0)	Q1(2^1)	Q2(2^2)	Q3(2^3)
12	0	0	1	1
13 (Reset)	0	0	0	0
(Q_{i-1}, Q_i)	(d)	(a)	(c)	(b)

Figure 15 mod-13 asynchronous counter state transition

The counter resets when the clock pulse hits 13, so there is a process or load from inside the counter to prevent 1010 (12) from becoming 1011 (13).

State (Clock)	Q0 (2^0)	Q1(2^1)	Q2(2^2)	Q3(2^3)
12	0 prevent Q0	0	1 force Q2 to fall	1 Q3 will be 0 if
13 (Reset)	0 from rising	0	0 to 0	0 Q2 changed
(Q_{i-1}, Q_i)	(d) to 1	(a)	(c)	(b)

Figure 16 mod-13 asynchronous counter state transition from State 12 to State 13(reset)

Thus, to realize a mod-13 asynchronous counter, this circuit in Figure 15 is configured.

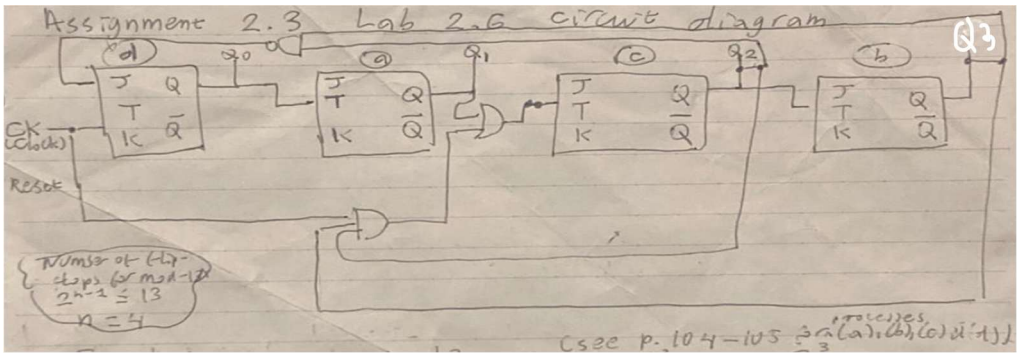


Figure 17 mod-13 asynchronous counter

FF0 undergoes process (d), FF1 goes through process (a), FF2 goes through process (c), and FF3 goes through process (b) of the mod-K asynchronous counter transition from state of K-1 to K (reset or 0, with K=13.

Waveform (in separate assignment paper)

The waveform is in an assignment paper to draw waveforms for assignment 2.3 which attached to the report and I marked as Figure 18. The waveforms are of the Clock, outputs Q0, Q1, Q2, and Q3, J0, the J input of the JK flip-flop outputting Q0, and T2,

the input T of the JK flip-flop outputting Q2. The arrows show the causal relationship between the waveforms.

The counter resets when the clock pulse reaches 13. As there is reset during the 13th clock pulse, the values of Q0, Q1, Q2, Q3, J0, and J2 all remain the same during the 14th clock pulse and so on.

J0 has a load to prevent Q0 to rise to 1. While T2 has a load (to force Q2 (to fall to 1?)?).

B) Synchronous counter configuration

An asynchronous counter uses the previous stage flip-flop output as a next stage clock, while a counter that supplies the same clock to all flip-flops and operates synchronized to each flip-flop is called a synchronous counter, and is considered a single synchronous sequential circuit.

Assignment 2.4 Mod-K synchronous counter

A mod-K synchronous counter sequentially transitions K states from 0 to K-1 according to a clock. A synchronous counter is a special sequential circuit with no input, in which a state transition function is a function only of internal state q.

Lab 2.7 mod-6 synchronous counter creation and operation explanation and confirmation

Three flip-flops are required to express six states from 0 to 5. Conforming to the counter concept, three flip-flops are made to correspond to each of the binary digits Q0, Q1, Q2, and six states are expressed by (Q0, Q1, Q2) = (0, 0, 0), (1, 0, 0), (0, 1, 0), (1, 1, 0), (0, 0, 1), (1, 0, 1). Looking at Q2 as the highest-order bit and Q0 as the lowest-order bit, the six states each correspond sequentially to the values from 0 to 5. Doing so enables a state transition table, as shown in Figure 19.

State	Present value			Next value		
	Q_0	Q_1	Q_2	Q'_0	Q'_1	Q'_2
0	0	0	0	1	0	0
1	1	0	0	0	1	0
2	0	1	0	1	1	0
3	1	1	0	0	0	1
4	0	0	1	1	0	1
5	1	0	1	0	0	0

Figure 20 mod-6 synchronous counter state transition state

Here, consider expressing Q'0 as much as possible as a simple logic function of Q0, Q1, Q2. When the relationship between Q'0 and Q0, Q1, and Q2 is shown in a Karnaugh map, which is a type of truth table, it appears as in Figure 19 (a). The places in the figure marked with * indicate where such states are not present, and a logic function

value of 1 or even 0 would be acceptable. Here, the value of * is decided such that the section enclosed by a diagonal line in Figure 20(a) becomes 1, and other sections become 0. Doing so implies that in the relationship in Figure 20(a), the logic function is expressed by $\overline{Q0}$. This implies that $\overline{Q0}$ can be used as a logic function providing Q'0. Similarly, when a Karnaugh map is created for Q'1 and Q'2 as in Figure 20(b) and (c), and the value of * is properly determined such that the form of the logic function is simplified, we ultimately find the following logic functions.

Q'0 = $\overline{Q0}$

Q'1 = $\overline{Q0} \cdot Q1 + Q0 \cdot \overline{Q1} \cdot \overline{Q2}$

Q'2 = Q0 · Q1 + $\overline{Q0}$ · Q2

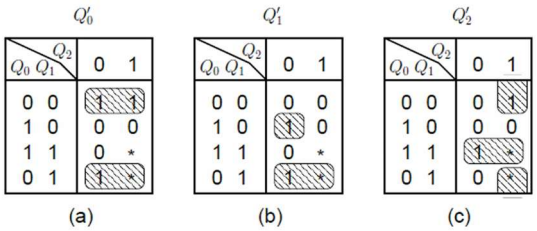


Figure 20 Karnaugh Map of Mod-6 Counter

Therefore, if a logic circuit is configured for each flip-flop such that the values Q'0, Q'1, and Q'2 are set according to the clock pulse, the three flip-flops operate as a mod-6 synchronous counter. The circuit diagram is in Figure 20 below.

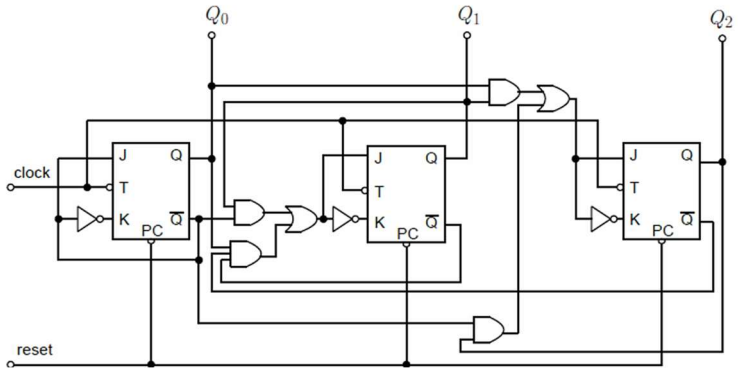


Figure 21 mod-6 synchronous counter

Lab 2.8 mod-5 synchronous counter creation and operation confirmation

Experimental details

We referred to above explanation in Lab 2.7 to design a mod -5 synchronous counter, configured the training device, and confirm its operation

Mod-5 synchronous counter design process

Three flip-flops are required to express five states from 0 to 4. Conforming to the counter concept, three flip-flops are made to correspond to each of the binary digits Q0, Q1, Q2, and five states are expressed by (Q0, Q1, Q2) = (0, 0, 0), (1, 0, 0), (0, 1, 0), (1, 1, 0), (0, 0, 1),). Looking at Q2 as the highest-order bit and Q0 as the lowest-order bit, the six states each correspond sequentially to the values from 0 to 5. Doing so enables a state transition table in the answer sheet for assignment 2.4 attached to the report.

We expressed Q'0, Q'1 and Q'2 each as a simple logic function of Q0, Q1, Q2. The relationship between Q'0 and Q'0 and Q0, Q1, and Q2 is shown in a Karnaugh map in the answer sheet for assignment 2.4 attached to the report. The places in the figure marked with * indicate where such states are not present, and a logic function value of 1 or even 0 would be acceptable. The value of * is properly determined such that the form of the logic function is simplified, we ultimately find the following logic functions.

From the Karnaugh map it can be determined:

Q'0 = $(\overline{Q0}) \cdot (\overline{Q2})$

Q'1 = Q1$\oplus$Q0 (where $\oplus$ denotes XOR operation)

Q'2 = $Q0 \cdot Q1$

The circuit diagram of mod 5-synchronous counter using 3 JK flip-flops is attached to the report and labelled as Figure 22.

Closing

From this lab we learned the basics of computer logic and memory operation.

In the first week, we learned to configure logic gates, combinational gates, decoder, and flip-flop on the training table and observe the waveform of flip-flops. Flip flop is a basic element of a sequential circuit and can be used to store memory as it can "remember" its previous state.

In the second week, we learned to configure asynchronous counters and synchronous counters an example of a sequential circuit.

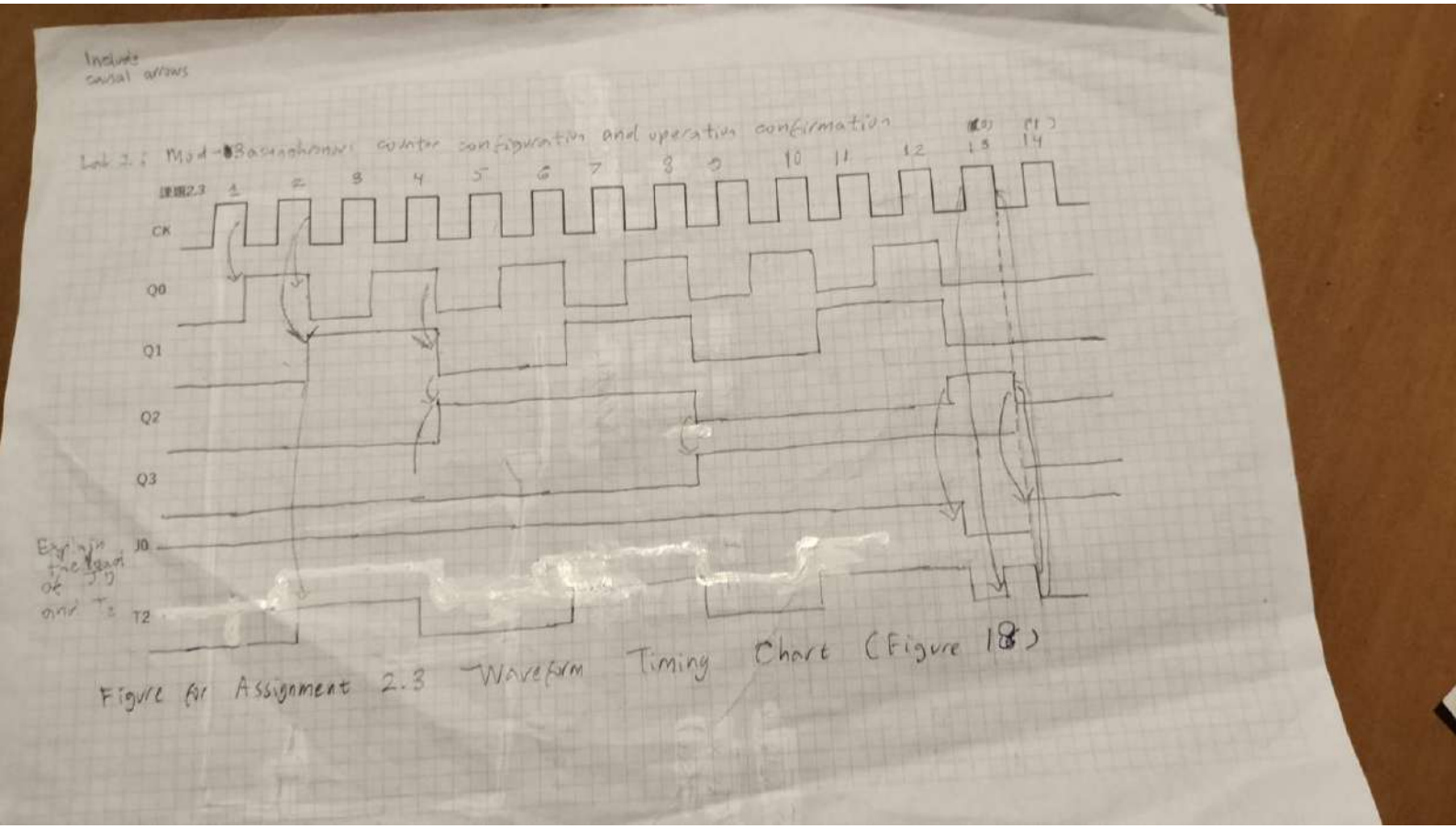


Figure for Assignment 2.3 Waveform Timing Chart (Figure 18)

Mod 5

Assignment 2.4

For assignment 2.4

Number of flip flops in mod-5

$2^{n-1} \leq 5$

$n = 3$

Reset at

状態遷移表

状態	現在の値 Q0	Q1	Q2	次の値 Q0'	Q1'	Q2'
0	0	0	0	1	0	0
1	1	0	0	0	1	0
2	0	1	0	1	1	0
3	1	1	0	0	0	1
4	0	0	1	0	0	0

カルノー図

Q0'

Q0 \ Q2	0	1
0	1	0
1	0	*
1	0	*
0	1	*

Q1'

Q1 \ Q2	0	1
0	0	*
1	1	*
1	0	*
0	1	*

Q2'

Q1 \ Q0	0	1
0	0	*
0	0	*
1	1	*
1	0	*

(論理式) $Q0' = \overline{Q_2} \cdot \overline{Q_0}$

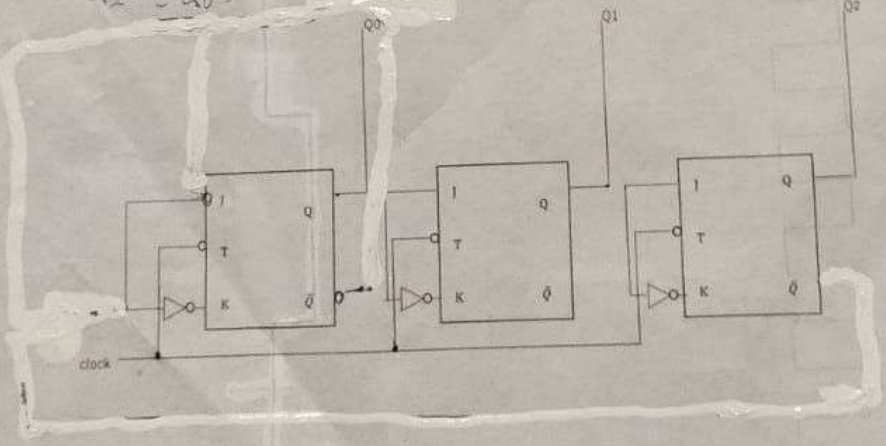
$Q1' = Q_1 \oplus Q_0$

⊕ is XOR

$Q2' =$

$Q_0' = \overline{Q_2 \cdot Q_0}$

回路図



(Circuit diagram is drawn on a separate page)

Assignment 2.4

Circuit Diagram

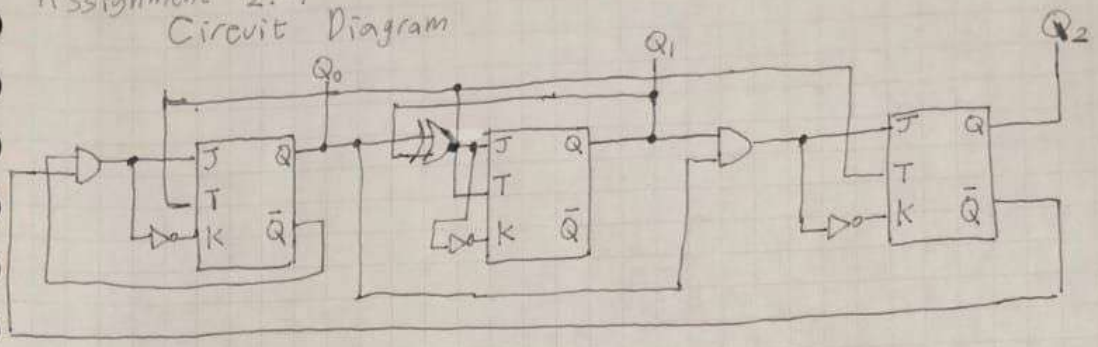


Figure 22 mod-5 synchronous counter